

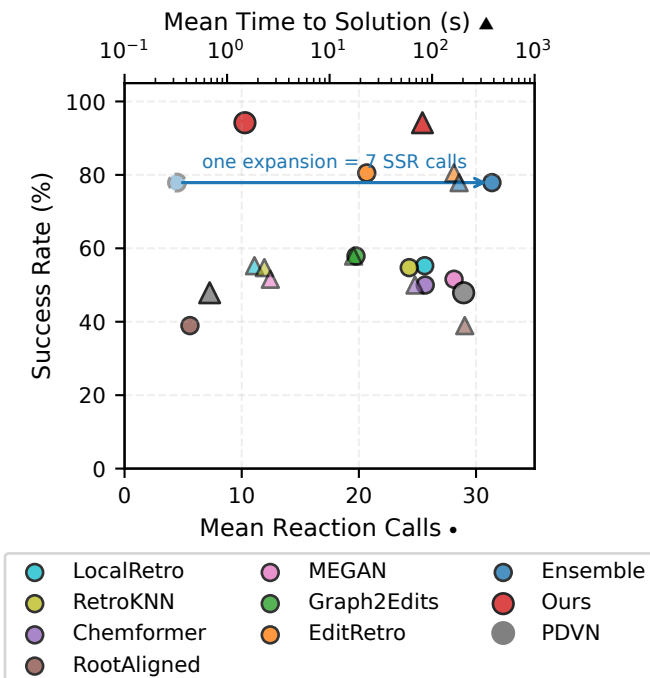


Figure r1: Efficiency-performance trade-offs between success rate and reaction-model calls to first solution (circles) versus time to first solution (triangles). One expansion in ensemble refers to 7 SSR calls.

Table r1: Sensitivity analysis of scaffold clustering similarity threshold on Retro*-190 dataset of RetrOrchestrator.

δ	Success Rate $\uparrow$	Avg. Wallclock Time (s)	Avg. No. of Expansions	Avg. No. of Unique Routes	Avg. Length of Routes	Avg. Util. Ratio $\uparrow$	Avg. SCScore $\uparrow$
0.8	71.5789%	200.2550	16.8603	1.2843	9.9771	56.3018%	3.9045
0.9	90.6250%	187.4097	10.3333	1.1829	7.0412	61.8465%	2.8474
0.95	94.2105%	118.9155	15.3473	1.7929	9.0075	76.4002%	3.9702
0.999	73.6842%	259.3644	14.3357	1.4095	10.6757	64.0259%	3.9151

Table r2: Performance comparison on Retro*-190 dataset of RetrOrchestrator trained with different clustering strategies. Molecule exact match corresponds to GiGPO setting in table 1 of the submitted paper.

Clustering Level	Similarity	Success Rate $\uparrow$	Avg. Wallclock Time (s)	Avg. No. of Expansions	Avg. No. of Unique Routes	Avg. Length of Routes	Avg. Util. Ratio $\uparrow$	Avg. SCScore $\uparrow$
Scaffold	0.95	94.2105%	118.9155	15.3473	1.7929	9.0075	76.4002%	3.9702
	0.999	73.6842%	259.3644	14.3357	1.4095	10.6757	64.0259%	3.9151
Molecule	0.999	74.2105%	192.7996	14.1348	2.9737	9.3050	65.8304%	3.9272
Search Graph	0.95	88.4211%	147.1972	11.2373	1.7619	9.7520	72.2474%	3.9727

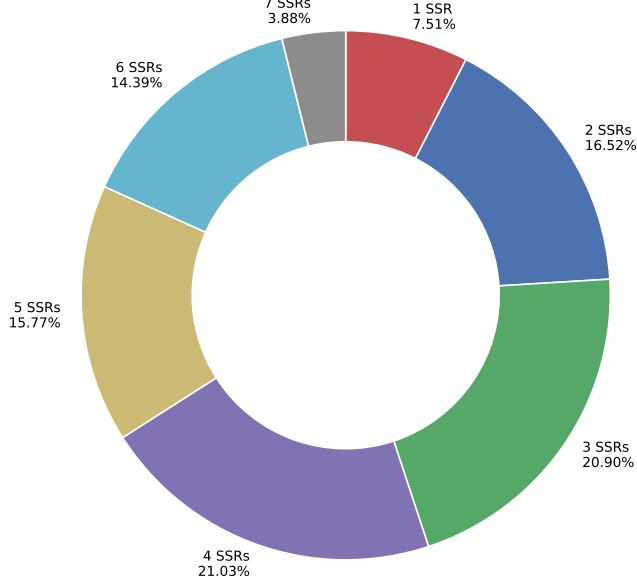


Figure r2: Breakdown of distinct SSRs in successful routes.

Table r3: Performance comparison on Retro*-190 dataset of RetrOrchestrator trained with different curriculum settings.

Setting	Success Rate $\uparrow$	Avg. Wallclock Time (s)	Avg. No. of Expansions	Avg. No. of Unique Routes	Avg. Length of Routes	Avg. Util. Ratio $\uparrow$	Avg. SCScore $\uparrow$
Stage 1 Only	65.4255%	219.7301	13.0000	1.9574	8.5610	65.8537%	3.9105
Stage 2 Only	72.2222%	90.0011	9.2614	1.9200	8.5124	78.8360%	3.9702
Cut off at SCScore=3.5	74.2647%	261.1774	13.3267	1.4605	7.9279	65.2303%	3.9078
Cut off at SCScore=4.0	94.2105%	118.9155	15.3473	1.7929	9.0075	76.4002%	3.9702
Cut off at SCScore=4.5	72.1053%	278.9027	13.0000	1.3302	8.9007	69.5677%	3.9301

Table r4: Performance comparison on Retro*-190 dataset of RetrOrchestrator trained with different weighting scheme between A_i^E and $A_{i,t}^S$. A_i^E only corresponds to GRPO setting in the paper.

Setting	Success Rate $\uparrow$	Avg. Wallclock Time (s)	Avg. No. of Expansions	Avg. No. of Unique Routes	Avg. Length of Routes	Avg. Util. Ratio $\uparrow$	Avg. SCScore $\uparrow$
A_i^E only	74.2105%	192.7996	14.1348	2.9737	9.3050	65.8304%	3.9272
$A_{i,t}^S$ only	72.1053%	278.9027	13.0000	1.3241	9.1469	60.5278%	3.9301
$A_i^E + A_{i,t}^S$	94.2105%	118.9155	15.3473	1.7929	9.0075	76.4002%	3.9702
$A_i^E + 5A_{i,t}^S$	60.0000%	261.6746	17.8333	1.0698	9.1957	49.6311%	3.9435

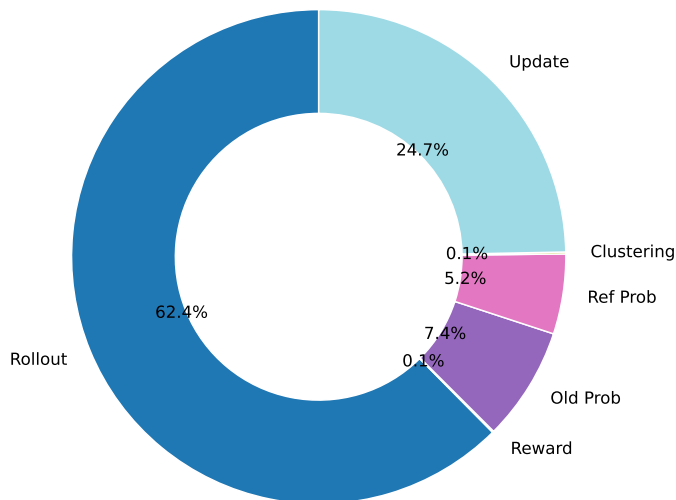


Figure r3: Training cost break down for each training step. Rollout contains both LLM inference, SSR model calls, and potential retrosynthesis environment overhead. Clustering includes the scaffold extraction process and scaffold-based step-level advantage calculation process.

Table r5: Performance comparison of static planner Retro* and RetrOrchestrator on larger, out-of-distribution PDB-600 dataset.

Planner	Setting	Success Rate $\uparrow$	Avg. Wallclock Time (s)	Avg. No. of Expansions	Avg. No. of Unique Routes	Avg. Length of Routes	Avg. Util. Ratio $\uparrow$	Avg. SCScore $\uparrow$
Retro*	LocalRetro	43.6667%	37.5267	15.3605	1.2060	4.0938	27.3817%	3.2392
	RetroKNN	46.3333%	29.7655	18.8381	1.2429	4.2463	22.1898%	3.3317
	Chemformer	42.5000%	69.3972	13.1136	1.2051	4.8476	31.4903%	3.2373
	RootAligned	53.3333%	129.0730	4.7289	1.5845	3.0456	52.9986%	3.7929
	MEGAN	53.8333%	37.6256	17.3868	1.2822	6.1130	28.3225%	3.8533
	Graph2Edits	57.6667%	31.4151	16.7231	1.2378	8.1000	28.6212%	3.8978
	EditRetro	59.6667%	9.9723	21.8405	1.0859	8.0631	30.8621%	3.5473
	Ensemble	47.3333%	105.1823	4.4902	2.1013	3.7571	58.1338%	3.7847
RetrOrchestrator		68.8333%	210.2934	8.5190	1.4237	4.9524	49.1828%	3.6453

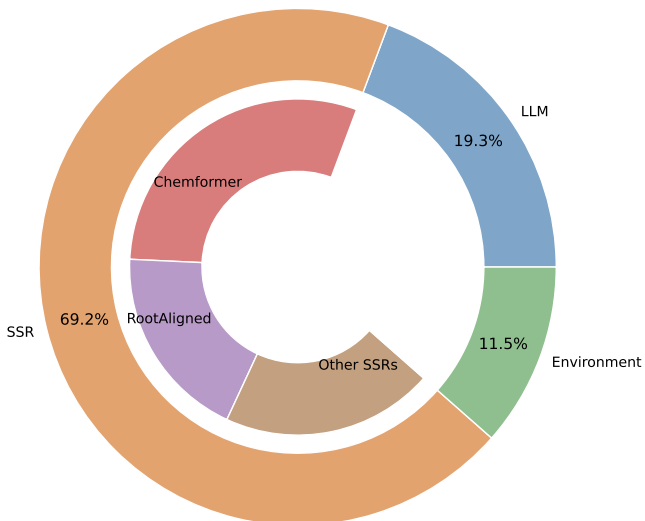


Figure r4: Rollout/inference runtime breakdown. Environment includes time spent on maintaining the search graph, checking commercial availability and other potential overhead.

Table r6: Performance comparison between RetrOrchestrator and PDVN under different expansion budget on Retro*-190 dataset.

Expansion Budget	Method	Success Rate $\uparrow$
10	PDVN	47.89%
	Ours	94.21%
50	PDVN	93.68%
	Ours	96.84%

Table r7: Performance comparison on Retro*-190 dataset of RetrOrchestrator against a non-LLM molecule-model router.

Setting	Success Rate $\uparrow$	Avg. Wallclock Time (s)	Avg. No. of Expansions	Avg. No. of Unique Routes	Avg. Length of Routes	Avg. Util. Ratio $\uparrow$	Avg. SCScore $\uparrow$
non-LLM Router	82.6316%	78.7745	22.2439	1.2166	7.0553	23.4125%	3.8385
Ours	94.2105%	118.9155	15.3473	1.7929	9.0075	76.4002%	3.9702

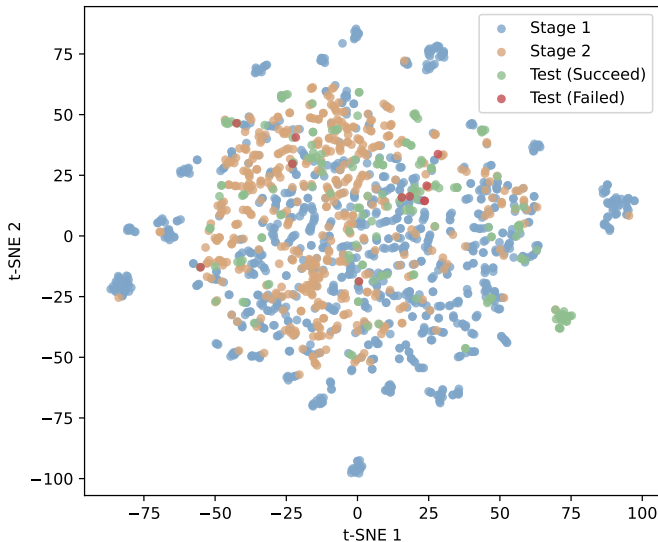


Figure r5: Distribution of Stage 1, Stage 2 training molecules as well as testing molecules from Retro*-190.

Table r8: Association of the number of predictions action dimension with dynamic search stage, model identity, and selected molecular features. The learned policy depends on both search stage and molecular structure, while model identity has a comparatively modest effect size.

Variable	Type	Metric	Value
Step	Dynamic context	Spearman ρ	0.211
Model identity	Static action identity	η^2	0.017
Hall-Kier alpha	Molecular feature	Spearman ρ	0.297
fr_piperdine	Molecular feature	Spearman ρ	0.274
Balaban's J	Molecular feature	Spearman ρ	0.262
Rotatable-bond fraction	Molecular feature	Spearman ρ	0.158
TPSA	Molecular feature	Spearman ρ	-0.347
fr_lactam	Molecular feature	Spearman ρ	-0.398
n_rings_size.4	Molecular feature	Spearman ρ	-0.398
fr_thiophene	Molecular feature	Spearman ρ	-0.354
fr_amide	Molecular feature	Spearman ρ	-0.329
fr_pyridine	Molecular feature	Spearman ρ	-0.315

Table r9: Chemistry-facing interpretation of the learned routing policy. We report motif-conditioned model preference using the increase in model call share when the motif/context is present (Δ share), together with molecule-level dominant-model consistency for sufficiently supported subgroups.

Motif / context	Preferred SSR	Δ call share	# mols	Dominant-model fraction
Piperidine motif (fr.piperidine >0)	EditRetro	+0.118	31	0.581
Chiral molecules (n.chiral.centers >0)	EditRetro	+0.039	105	0.457
Amide-containing molecules (fr.amide >0)	EditRetro	+0.031	47	0.574
Heteroatom-rich molecules	Chemformer	+0.086	122	0.393
Benzene-rich scaffolds (fr.benzene >0)	Chemformer	+0.047	88	0.443
High-TPSA molecules	Chemformer	+0.045	119	0.471
Pyridine-containing molecules (fr.pyridine >0)	Chemformer	+0.043	35	0.657 [†]
Fused-ring topology (n.fused.ring.pairs >0)	LocalRetro	+0.009	61	0.443 [†]

[†] Molecule-level dominant model for this subgroup is EditRetro, while the action-level Δ call-share increase is largest for the listed model. This indicates that call allocation and dominant-model identity need not coincide exactly.

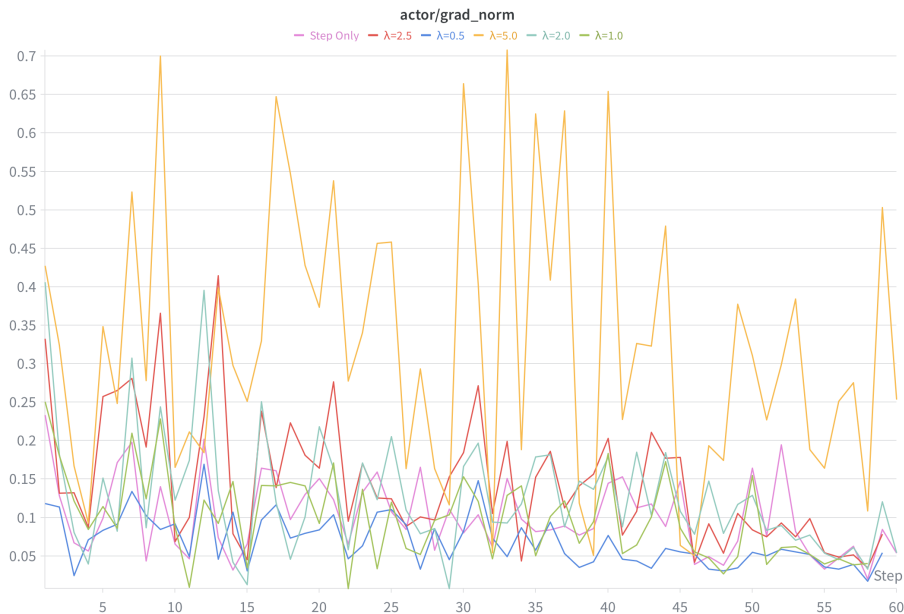


Figure r6: Per-step gradient norm gathered during ablation studies on weighting scheme.

Table r10: Chemistry-facing interpretation of selected descriptors associated with the action. RDKit fragment descriptors (**fr_**xxx) are SMARTS-based motif counts.

Feature	Interpretation
Step	Search-stage index; larger values correspond to later surviving search states in the search trajectory.
Model identity	Which one-step retrosynthesis model (SSR) is selected for the action.
Hall-Kier alpha	Topological descriptor related to molecular shape, heteroatom substitution, and overall graph complexity.
Balaban's J	Graph-topological complexity / connectivity index reflecting branching and cyclic connectivity patterns.
Rotatable-bond fraction	Fraction of bonds that are rotatable; proxy for conformational flexibility.
Ring-bond fraction	Fraction of molecular bonds associated with ring structure; proxy for ring-dominated topology.
TPSA	Topological polar surface area; proxy for polarity and heteroatom-rich functionality.
HBD	Number of hydrogen-bond donor groups; typically hydroxyls, amines, etc.
HBA	Number of hydrogen-bond acceptor atoms; typically heteroatoms such as O, N, S in accepting environments.
n_hetero_atoms	Number of non-carbon/non-hydrogen atoms (e.g., O, N, S, halogens).
hetero_atom_fraction	Fraction of atoms that are heteroatoms; a normalized measure of heteroatom richness.
n_to_c_ratio	Ratio of nitrogen atoms to carbon atoms.
n_O	Number of oxygen atoms.
n_N	Number of nitrogen atoms.
n_S	Number of sulfur atoms.
n_double_bonds	Number of double bonds.
n_rings	Total number of rings.
n_rings_size_3	Number of 3-membered rings.
n_rings_size_4	Number of 4-membered rings.
n_rings_size_5	Number of 5-membered rings.
n_fused_ring_pairs	Number of fused ring pairs; proxy for fused polycyclic topology.
n_chiral_centers	Number of stereogenic center; proxy for stereochemical complexity.
fr_Al.OH	Count of aliphatic alcohol motifs.
fr_amide	Count of amide motifs.
fr_ether	Count of ether motifs.
fr_lactam	Count of lactam (cyclic amide) motifs.
fr_piperidine	Count of piperidine-like saturated N-heterocycle motifs.
fr_pyridine	Count of pyridine-like aromatic N-heterocycle motifs.
fr_thiophene	Count of thiophene ring motifs.
fr_benzene	Count of benzene motif; proxy for aromatic / benzene-rich scaffolds.
fr_aryl_methyl	Count of aryl-methyl substituent motif, i.e., a methyl attached to an aromatic system.

Table r11: Top-correlating molecule features for the number of predictions action dimension.

chemformer	editretro	graph2edits	localretro	megan	retroknn	root _a ligned
fr.lactam	tpsa	fr.lactam	fr.lactam	n.rings.size.4	fr.lactam	n.rings.size.3
n.rings.size.4	fr.amide	n.rings.size.4	n.rings.size.4	fr.lactam	n.rings.size.4	hba
fr.thiophene	hall_kier_alpha	n.fused_ring.pairs	n.fused_ring.pairs	hbd	fr.thiophene	fr.AL_OH
fr.amide	fr.pyridine	fr.thiophene	fr.piperidine	n.N	n.S	n.hetero.atoms
n.fused_ring.pairs	fr.ether	n.S	hbd	tpsa	balaban_j	n.rings.size.4
n.rings	n.fused_ring.pairs	fr.pyridine	ring.bond.fraction	fr.thiophene	n.fused_ring.pairs	fr.lactam
n.N	fr.thiophene	tpsa	rotatable.bond.fraction	n.S	fr.amide	fr.thiophene
balaban_j	n.rings.size.4	fr.piperidine	n.S	n.to.c.ratio	ring.bond.fraction	hetero.atom.fraction
tpsa	fr.lactam	n.rings.size.5	fr.thiophene	hba	n.rings.size.5	n.S
fr.pyridine	fr.piperidine	n.double.bonds	n.rings	n.rings	n.rings	n.O

Table r12: Average gradient norms under different advantage-weighting settings. Notably, higher gradient norms indicate increased training instability.

Setting	$\lambda = 0.5$	$\lambda = 1.0$	$\lambda = 2.0$	$\lambda = 2.5$	$\lambda = 5.0$	Step Only
Gradient norm	0.06939	0.09891	0.12817	0.14672	0.32397	0.10129